

Windows Server 2016 — Comprehensive Implementation Summary

1. Server Identity & Environment

Parameter	Value
Server Hostname	WINSERVER
Server IP Address	30.0.0.20
Operating System	Windows Server 2016
Domain Name	uni.local
Forest Functional Level	Windows Server 2016
Domain Functional Level	Windows Server 2016
Network (Management)	192.168.183.128 (PNETLab access)

The server was deployed inside a virtualized lab environment running on PNETLab/QEMU, reachable at 192.168.183.128, and serves as the centralized Domain Controller, DNS Server, File Server, and Group Policy authority for the entire enterprise network.

2. Roles & Features Installed

The following roles and features were installed through **Server Manager → Add Roles and Features**:

Role / Feature	Purpose
Active Directory Domain Services (AD DS)	Centralized identity, authentication, and policy management
DNS Server	Internal name resolution and AD service location
File and Storage Services	Centralized file sharing across departments
NPAS (Network Policy and Access Services)	Network access control
Windows Server Backup	System state backup and disaster recovery

All installed roles are visible in the Server Manager Dashboard, each showing a green **Manageability** indicator confirming no critical errors.

3. Active Directory Domain Services (AD DS)

3.1 Domain Promotion

After installing the AD DS role, the server was promoted to a **Domain Controller** for the new forest `uni.local`. During promotion:

- **Forest and Domain Functional Level** were both set to Windows Server 2016.
- The **DNS Server** option was enabled during promotion so the DC also acts as the primary DNS server.
- A **DSRM (Directory Services Restore Mode)** password was configured for recovery scenarios.
- The **NetBIOS domain name** was confirmed as UNI.

3.2 Organizational Unit (OU) Structure

Three top-level **Organizational Units** were created directly under the `Uni.local` domain root to mirror the university's department structure:

OU Name	Target Population
<code>uni_Student</code>	Student users and their workstations
<code>uni_Staff</code>	Faculty/staff users and their workstations
<code>uni_Management</code>	Administrative/management users and workstations
<code>uni_Groups</code>	Security groups for RBAC
<code>NetworkAdmin</code>	Network administration accounts

This OU design is the foundation of the entire security model — every Group Policy, every permission assignment, and every access control decision flows from which OU a user or computer object lives in.

3.3 User Accounts

User accounts were created inside their respective OUs. Each account was configured with:

- A **username** following a naming convention (e.g., Student 1, Student 2, Student 3).
- An **initial password** set by the administrator.
- A **Description** field populated with the VLAN/department tag (e.g., "Vlan 10 | Students", "Vlan 40 | Students") for easy identification and auditing.
- **Domain authentication**, allowing login from any domain-joined workstation.

Confirmed user accounts visible in `uni_Student`:

- Student 1 — Vlan 10 | Students
- Student 2 — Vlan 10 | Students

- Student 3 — Vlan 40 | Students

3.4 Computer Accounts (Domain-Joined Workstations)

The following Windows 10 client machines were joined to the `uni.local` domain and their computer objects were moved into the correct OUs:

Computer Name	Description	OU
DESKTOP-M65LJ28	Management workstation	uni_Student / Management
STUDENT	win_10.1 (Student)	uni_Student
STUDENT-TWO-V10	win_10.2 (Student)	uni_Student

Each machine was joined by going to **This PC** → **Properties** → **Change Settings** → **Domain** → **uni.local**, providing administrator credentials, and rebooting. After joining, the computer objects were moved from the default `Computers` container into the appropriate department OU so that the correct GPOs apply to them.

4. Security Groups (RBAC)

Three **Global Security Groups** were created in the `uni_Groups` OU to implement **Role-Based Access Control (RBAC)** across the file server and all permission assignments:

Group Name	Members
Students-GRP	All student user accounts
Staff-GRP	All staff/faculty user accounts
Management-GRP	All management/admin user accounts

Permissions are assigned to **groups, not individual users**. This means adding or removing a user from a department only requires adding/removing them from the group — access is updated everywhere automatically.

5. DNS Server Configuration

5.1 Forward Lookup Zone — uni.local

The Forward Lookup Zone `uni.local` was created automatically during AD DS promotion. The following **Host (A) records** were configured, mapping hostnames to IP addresses:

Hostname	IP Address	Type	Notes
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Hostname	IP Address	Type	Notes
winserver	30.0.0.20	Host (A) — static	Domain Controller / DNS Server
Student	192.168.10.19	Host (A)	Client workstation
Student-one-Vlan10	192.168.10.19	Host (A)	Student VLAN 10
Student-two-V10	192.168.10.18	Host (A)	Student VLAN 10
DESKTOP-M65LJ28	192.168.40.19	Host (A)	Management workstation
Staff-1-V30	192.168.30.29	Host (A)	Staff VLAN 30
Staff-2-V30	192.168.30.18	Host (A)	Staff VLAN 30
Staff-3-V60	192.168.60.19	Host (A)	Staff VLAN 60
Management-one-V20	192.168.20.19	Host (A)	Management VLAN 20
Management-2-V20	192.168.20.18	Host (A)	Management VLAN 20
Management-3-V50	192.168.50.29	Host (A)	Management VLAN 50

In addition, the **SOA (Start of Authority)** and **NS (Name Server)** records were automatically created and point to `winserver.uni.local`.

5.2 Reverse Lookup Zone — 30.in-addr.arpa

A Reverse Lookup Zone `30.in-addr.arpa` was manually created to allow **IP-to-hostname** resolution. A **Pointer (PTR) record** was configured mapping `30.0.0.20` → `WinServer.Uni.local` (static). SOA and NS records were also automatically created in this zone.

5.3 Client DNS Configuration

Every client machine was configured to use the server's IP (`30.0.0.20`) as its **Primary DNS Server** in the network adapter's IPv4 properties, ensuring all domain name lookups are handled by the domain controller.

6. Centralized File Server

6.1 Shared Folder Structure

A parent shared folder named "**Uni Shared Folder**" was created on the server (located at `C:\Users\Administrator\Desktop\Uni Shared Folder`). Three department-specific sub-folders were created inside it:

```
Uni Shared Folder\
├── Shared-Students\    (created 6/27/2026 1:58 PM)
├── Shared-Staff\       (created 6/27/2026 1:58 PM)
└── Shared-Management\ (created 6/27/2026 1:58 PM)
```

The parent folder was shared on the network as \\30.0.0.20\Uni Shared Folder, making all three sub-folders accessible to domain clients.

6.2 NTFS (Security) Permissions

NTFS permissions were configured on the folder via **Properties** → **Security tab** following the **Principle of Least Privilege**:

Principal	Permissions Granted
Authenticated Users	Read & Execute, List Folder Contents, Read
SYSTEM	Full Control
Administrator	Full Control
Administrators (UNI\Administrators)	Full Control

For each department sub-folder (Shared-Students, Shared-Staff, Shared-Management), the corresponding security group (Students-GRP, Staff-GRP, Management-GRP) was granted access while inheritance from the parent was broken to ensure **strict isolation between departments**.

6.3 Share (Network) Permissions

Configured via **Properties** → **Sharing** → **Advanced Sharing** → **Permissions**:

Principal	Permissions
Authenticated Users	Full Control ✓, Change ✓, Read ✓

The combination of Share Permissions (allowing all authenticated users through the network share) and NTFS Permissions (restricting access to the correct department group at the file-system level) provides the real enforcement — NTFS is always the stricter layer.

6.4 Access Control Verification

A **Student account** logged onto a client machine and navigated to \\30.0.0.20\Uni Shared Folder:

- **Shared-Students** → Access granted ✓
- **Shared-Staff** → Received the error: *"Windows cannot access \\30.0.0.20\Uni Shared Folder\Shared-Staff. You do not have permission to access..."* ✗

This confirmed that NTFS permissions correctly enforced cross-department access isolation.

7. Group Policy Objects (GPO)

All GPOs were created and linked through **Server Manager** → **Tools** → **Group Policy Management**. The procedure for every policy was:

1. Right-click the target OU → "**Create a GPO in this domain, and Link it here...**"
2. Name the GPO, then right-click it → **Edit** to open the Group Policy Management Editor.
3. Navigate to the appropriate node (User Configuration or Computer Configuration) and enable the required setting.

7.1 Students GPO — uni_Student OU (Most Restrictive)

14 GPOs were created and linked to uni_Student, each targeting a specific restriction:

#	GPO Name	Policy Path	Setting Enabled
1	Close Control Panel	User Config → Admin Templates → Control Panel	Prohibit access to Control Panel and PC settings
2	Prevent Registry editor	User Config → Admin Templates → System	Prevent access to registry editing tools
3	Prevent access to Task Manager	User Config → Admin Templates → System → Ctrl+Alt+Del Options	Remove Task Manager
4	No Wallpaper	User Config → Admin Templates → Desktop → Desktop	Desktop Wallpaper (path left blank)
5	USB lock	Computer Config → Admin Templates → System → Removable Storage Access	All Removable Storage classes: Deny all access
6	Hide Drivers	User Config → Admin Templates → Windows Components → File Explorer	Hide these specified drives in My Computer
7	Can't change Password	User Config → Admin Templates → System → Ctrl+Alt+Del Options	Remove Change Password
8	Prevent Access to shutdown	User Config → Admin Templates → System → Ctrl+Alt+Del Options	Remove Shut Down, Restart, Sleep, and Hibernate
9	Hide Network	User Config → Admin Templates → Network → Network Connections	Prohibit access to properties of a LAN connection / hide Network icon
10	Logon Banner	Computer Config → Policies → Windows Settings → Security Settings → Local Policies → Security Options	Interactive logon: Message title + Message text
11	Prevent Access of date and time	User Config → Admin Templates → Control Panel	Hide Date and Time icon / prevent modification
12	CMD	User Config → Admin Templates → System	Prevent access to the command prompt
13	Logon Hours	Enforced both via per-user AD account settings AND a linked GPO	Restricts login to authorized working hours
14	Close PowerShell	User Config → Admin Templates → System	Don't run specified Windows applications (powershell.exe)

GPO Enforcement Confirmed: A student account on a domain-joined machine attempted to open a restricted application and received the pop-up: *"Restrictions — This operation has been cancelled due to restrictions in effect on this computer. Please contact your system administrator."*

7.2 Staff GPO — uni_Staff OU (Moderate Restrictions)

8 GPOs linked to uni_Staff, balancing usability with security:

#	GPO Name	Setting
1	Can't change Password	Remove Change Password
2	Close PowerShell	Block powershell.exe execution
3	Close Control Panel	Prohibit access to Control Panel
4	Logon Hours	Restrict login to authorized hours
5	Prevent Registry editor	Prevent registry editing tools
6	Prevent access to Task Manager	Remove Task Manager
7	Hide Network	Hide Network settings
8	Logon Banner	Security warning at login screen

7.3 Management GPO — uni_Management OU (Administrative Restrictions)

7 GPOs linked to uni_Management:

#	GPO Name	Setting
1	Can't change Password	Remove Change Password
2	Hide Network	Hide Network settings
3	Logon Banner	Security warning at login screen
4	USB lock	Deny all removable storage access
5	Prevent Registry editor	Prevent registry editing tools
6	Prevent Access of date and time	Prevent date/time modification
7	Logon Hours	Restrict login to authorized hours

8. Password Policy

Configured under **Default Domain Policy** → **Computer Configuration** → **Windows Settings** → **Security Settings** → **Account Policies** → **Password Policy**:

Setting	Purpose
Enforce Password History	Prevents reusing recently used passwords
Minimum Password Length	Enforces a minimum character count

Setting	Purpose
Password Must Meet Complexity Requirements	Requires uppercase, lowercase, digits, and symbols
Maximum Password Age	Forces periodic password changes

9. Account Lockout Policy

Configured under **Default Domain Policy** → **Account Policies** → **Account Lockout Policy**:

Setting	Purpose
Account Lockout Threshold	Number of failed logon attempts before the account locks
Account Lockout Duration	How long the account stays locked before auto-reset
Reset Account Lockout Counter After	The time window for counting failed attempts

This policy directly mitigates **brute-force and password spraying attacks** by automatically locking accounts that receive too many wrong password attempts.

10. Security Logon Banner

Configured via **Computer Configuration** → **Policies** → **Windows Settings** → **Security Settings** → **Local Policies** → **Security Options**:

- **Interactive logon: Message title for users attempting to log on** → set to a warning title (e.g., "Authorized Users Only").
- **Interactive logon: Message text for users attempting to log on** → set to a full warning message notifying users that the system is monitored and restricted to authorized personnel.

This banner appears **before** the credential prompt on every domain-joined workstation, serving both as a legal notice and a security deterrent.

11. Logon Hours

Logon hours were configured **per individual user account** via: **Active Directory Users and Computers** → **User Properties** → **Account tab** → **Logon Hours button**

In the grid view (days of the week vs. hours of the day):

- **Blue (Logon Permitted)** blocks were selected for working hours (Sunday through Thursday, specific hour ranges).
- **White (Logon Denied)** blocks were left for off-hours (evenings, Friday, Saturday).

This prevents users from logging onto the domain outside authorized working hours. The "**Logon Hours**" GPO was also linked to each OU to enforce this at the policy level in addition to the per-user setting.

12. Windows Security Auditing

Advanced Audit Policies were enabled through **Group Policy Management Editor** → **Computer Configuration** → **Policies** → **Windows Settings** → **Security Settings** → **Advanced Audit Policy Configuration**:

Audit Category	What is Logged
Logon / Logoff	Every successful and failed logon/logoff event
Account Management	User account creation, deletion, modification; group changes
Object Access	File system access; file share access
Policy Change	Changes to the audit policy itself or security settings
Computer Account Management	Computer objects joining/leaving the domain

For file-system auditing to generate events, a **SACL (System Access Control List)** was also added to the shared folders via **Properties** → **Security** → **Advanced** → **Auditing**, specifying which access types (Read, Write, Delete) trigger audit log entries and for which users/groups.

13. Event Viewer — Security Log

The **Windows Event Viewer** (Server Manager → Tools → Event Viewer → Windows Logs → Security) captures all configured audit events. The Security log at the time of review contained **147,041 recorded events**.

Key Event IDs observed:

Event ID	Description
4624	An account was successfully logged on
4634	An account was logged off
4656	A handle to an object was requested (File System access)
5140	A network share object was accessed (File Share)

Each event record includes: the date/time, the source (Microsoft Windows security auditing), the subject (user who performed the action), the target resource, and the outcome (Success/Failure).

14. Remote Desktop Protocol (RDP)

RDP was enabled on the server via **This PC → Properties → Advanced System Settings → Remote tab**:

- Selected "**Allow remote connections to this computer**"
- Kept **Network Level Authentication (NLA)** enabled for pre-authentication security
- Added authorized administrator accounts via "**Select Users...**"

The Windows Firewall rule group "**Remote Desktop**" was automatically enabled on TCP port 3389, allowing inbound RDP sessions from authorized management machines.

15. Windows Server Backup

15.1 Feature Installation

The **Windows Server Backup** feature was installed through: **Server Manager → Add Roles and Features → Features → Windows Server Backup → Install**

15.2 Backup Job Execution

A manual backup job was executed using the **Backup Once Wizard** (wbadmin console → Actions → Backup Once):

Wizard Step	Choice Made
Backup Options	Different options (custom configuration)
Select Backup Configuration	Custom → System State selected
Specify Destination Type	Local drives
Select Backup Destination	Drive M:
Confirmation	Verified settings, clicked Backup

Backup progress was monitored live: the System state backup transferred approximately **7.17 GB** total, reaching completion successfully.

15.3 Backup Result

The **Windows Server Backup dashboard** (Local Backup view) confirmed:

Field	Value
Last Backup Status	Successful ✓
Last Backup Time	6/28/2026 9:12 PM
Next Backup Status	Not scheduled (one-time backup)

15.4 Backup Files

Navigating to the backup destination (`WinServer\Backup 2026-06-28 191233`) revealed **16 items**:

- `4b701312-0000-0000-0000-501f00000000....` — **Hard Disk Image File (7,843,840 KB ≈ 7.48 GB)**
- `BackupSpecs.xml` — Backup configuration descriptor
- Multiple `dae4765e-5421-4c8c-adb1-c8961e3a31f0....xml` — Component writer metadata documents (2 KB–5,239 KB each)

The Hard Disk Image File contains the full backup of the protected volumes and system state.

16. Network Topology Context

The Windows Server sits at the center of a multi-VLAN enterprise network built in PNETLab:

VLAN	Subnet	Department
VLAN 10	192.168.10.0/24	Students
VLAN 20	192.168.20.0/24	Management
VLAN 30	192.168.30.0/24	Staff
VLAN 40	192.168.40.0/24	(additional client segment)
VLAN 50	192.168.50.0/24	Management extension
VLAN 60	192.168.60.0/24	Staff extension
VLAN 10 (GW)	192.168.10.1	Gateway
Server subnet	30.0.0.0/24	Domain Controller

The network passes through a Gateway (192.168.10.1) with routing between VLANs managed by the network devices (switches and router visible in the PNETLab topology). The Windows Server at 30.0.0.20 is reachable from all VLANs for authentication, DNS, file access, and policy distribution.

17. Security Architecture Summary

The overall security posture implemented on this server follows four layers:

Layer 1 — Identity & Access (AD DS) Centralized authentication via Kerberos; all users must authenticate against the domain before accessing any resource. No local accounts are used for network resources.

Layer 2 — Endpoint Hardening (GPO) 14 policies on Students, 8 on Staff, 7 on Management — each progressively relaxed as privilege increases. Students have the most restricted desktop environment (no CMD, no PowerShell, no Control Panel, no USB, no registry). Staff and Management retain more usability but are still locked down on critical vectors (registry, task manager, PowerShell, USB for Management).

Layer 3 — Resource Access Control (NTFS + Share Permissions) Every department folder is siloed — a user in Students-GRP cannot read, write, or even list the contents of Shared-Staff or Shared-Management. Access Denied is enforced at the NTFS layer, which cannot be bypassed even with share-level access.

Layer 4 — Monitoring & Auditing (Audit Policy + Event Viewer) Every logon, logoff, account change, file access, and policy change is written to the Security event log. With 147,041+ events recorded, the audit trail provides full accountability and supports incident investigation.

This summary covers all configurations implemented on the Windows Server 2016 Domain Controller based on the live environment screenshots and implementation records.